IOT AND AI-BASED LANDSLIDE MONITORING AND EARLY WARNING SYSTEM FOR REAL-TIME RISK PREDICTION

A Project Report
submitted
in partial fulfilment of the requirements for the degree of

BACHELOR OF TECHNOLOGY

in
Civil Engineering
(Session: 2025-26)
by

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April, 2026

DECLARATION

I/We hereby declare that the work presented in this report entitled **“IoT and AI-Based Landslide Monitoring and Early Warning System for Real-Time Risk Prediction"**, was carried out by me/us. I/We have not submitted the matter embodied in this report for the award of any other degree or diploma of any other University or Institute.

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This is to certify that the Project Report entitled **"IoT and AI-Based Landslide Monitoring and Early Warning System for Real-Time Risk Prediction**" submitted by **Shashank Kumar Singh** in a partial fulfillment for the award of Bachelor's Degree in Civil Engineering for Major Project [BCE 753] as prescribed by Dr. APJ Abdul Kalam Technical University, Lucknow Uttar Pradesh represents the work carried out by students under my supervision. The project embodies result of original work and studies carried out by the students themselves and the contents of the project do not form the basis for the award of any other degree to the candidate or to anybody else.

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ABSTRACT

Landslides are among the most destructive natural disasters, posing a major threat to life, property, and critical infrastructure across India's hilly terrains. Every year, thousands of people are affected in states like Uttarakhand, Himachal Pradesh, Sikkim, and Kerala due to sudden slope failures triggered by heavy rainfall, deforestation, and unplanned construction. Conventional monitoring systems are expensive, manual, and lack real-time alert capabilities, leaving communities highly vulnerable. The monitoring of ground water changes is designed by using soil moisture sensor. To address this challenge, the Landslide Alert & Monitoring Program (L.A.M.P) has been developed — an AI-driven, IoT-based early warning system designed for continuous, intelligent, and cost-effective landslide risk prediction. The system deploys soil moisture sensors and 3-axis gyroscopes embedded 1.5–2 meters underground at multiple points in high-risk slopes. These sensors continuously record soil saturation, ground tilt, and rainfall intensity. The data are analyzed through a machine learning model that determines adynamic risk probability score, predicting landslide like Lihood well before slope failure occurs. When risk levels cross critical thresholds, the system automatically sends real-time alerts via mobile application, SMS, and local siren modules to residents and authorities, ensuring timely evacuation and rapid disaster response. This innovation directly supports national initiatives such as the National Landslide Risk Management Strategy (NDMA), the GSI Landslide Hazard Zonation Program, and the Digital India Mission, which emphasize smart, data-driven disaster management. By combining AI analytics, IoT sensing, and community awareness, L.A.M.P provides a scalable, low-cost, and field-ready solution that aligns with the vision of Atmanirbhar Bharat and promotes safer, sustainable, and resilient hill infrastructure across India

KEYWORDS: Soil Moisture Sensor, Gyroscopes, Conventional monitoring systems, Resilient hill infrastructure

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CHAPTER 1
INTRODUCTION

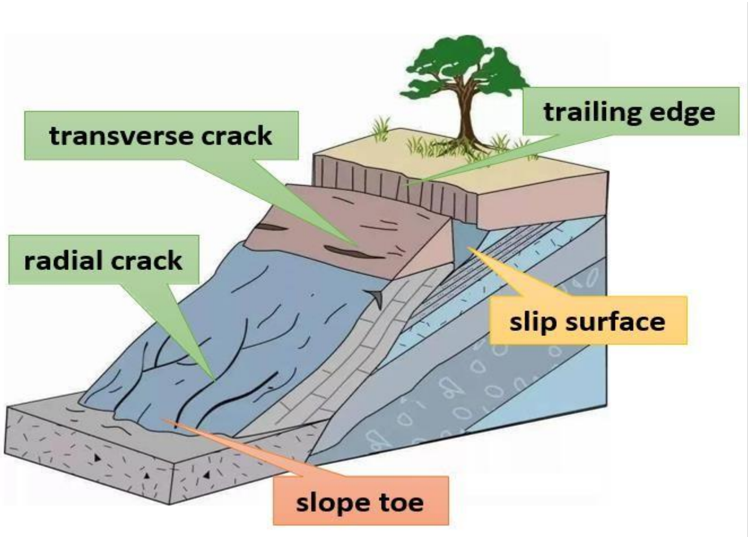
Landslides are among the most destructive natural disasters in India, causing extensive loss of life, property, and infrastructure every year. Regions such as Uttarakhand, Himachal Pradesh, Sikkim, Kerala, and the Northeastern states are particularly vulnerable due to steep slopes, heavy monsoon rainfall, deforestation, and unplanned construction. Traditional landslide monitoring methods, including manual inspections and periodic surveys, are often time- consuming, costly, and unable to provide real-time alerts, leaving communities highly exposed to sudden slope failures. To address these challenges, the Smart Landslide Alert & Monitoring Program (SMART L.A.M.P) proposes an AI-driven, IoT-based early warning system capable of continuous monitoring, intelligent prediction, and rapid alert dissemination. The system integrates soil moisture sensors and 3-axis gyroscopes embedded at critical depths in high-risk slopes. These devices continuously monitor soil saturation, ground movement, and rainfall intensity, feeding real-time data to a machine learning model that calculates a dynamic landslide probability score.

When the calculated risk exceeds defined thresholds, the system automatically sends real-time alerts to residents and authorities via mobile applications, SMS, and sirens, enabling timely evacuation and preventive action.

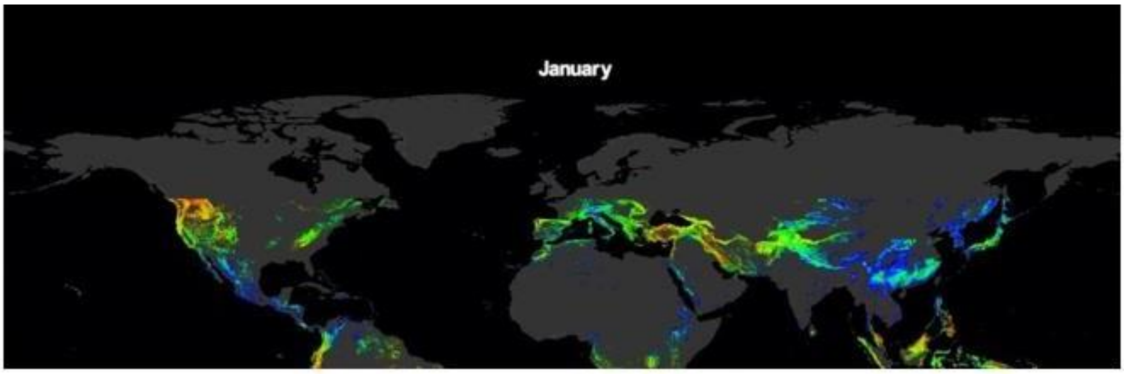
This project aligns directly with national initiatives, such as:

- National Landslide Risk Management Strategy (NDMA) – emphasizes early warning systems and disaster preparedness.
- GSI Landslide Hazard Zonation Program – supports slope monitoring and hazard mapping.
- Digital India Mission – promotes data-driven, technology-based governance and disaster management.
- Smart Cities Mission and Atmanirbhar Bharat – encourage indigenous, low-cost, scalable solutions for public safety.
- By combining AI analytics, IoT sensing, and community awareness, SMART L.A.M.P offers a low-cost, scalable, and field-ready solution to reduce casualties, protect property, and build resilient infrastructure in landslide-prone regions of India.

- The Challenge of Landslide Alert and Monitoring
- While many regions are predominantly known for flood risks, the eastern areas and adjacent hilly zones face a growing and under-addressed threat from landslides. Unlike large-scale mountain slides, these are often smaller yet highly destructive slope failures affecting road embankments, riverbanks, and natural escarpments. The region's unique geological and climatic conditions present a distinct set of challenges for implementing an effective landslide alert and monitoring program.
- The primary trigger is intense and prolonged monsoon rainfall, which saturates the soil profiles. Vulnerable slopes often consist of a mixture of silty clay and sand that loses shear strength drastically upon saturation. This hydrogeological process makes soil moisture a critical and direct indicator of impending slope instability.
- The challenges in addressing this problem are multifaceted:
- Data Scarcity: There is a significant lack of historical and real-time geotechnical data on slope stability, making it difficult to identify all high-risk zones and model failure thresholds accurately.
- Reactive Approach: Current practices are largely reactive, with interventions occurring only after slope failures have damaged infrastructure such as highways, roads, and residential areas.
- Resource Limitations: Deploying traditional monitoring equipment across multiple potential sites is economically unfeasible for local authorities.
- Public Awareness: Communities living near these slopes are often unaware of landslide risks, leading to construction and agricultural activities that further destabilize the terrain.
- Therefore, the central challenge is to develop a low-cost, reliable, and real-time monitoring system that can detect hydrological triggers specific to soil type and climatic conditions. Such a system would enable proactive measures, reduce casualties, protect infrastructure, and strengthen community resilience in landslide-prone areas.



The potential landslide activity by month area



- The landslide disaster is the first geological disaster and the damage is serious get over the last 15 years.

Fig.1.1. Landslide Real Monitoring

Need for the Project:

1. Landslides are a devastating natural hazard, causing significant loss of life, infrastructure damage, and economic setbacks, especially in vulnerable, rainfall-prone regions. Traditional geotechnical monitoring methods, such as manual inclinometers and piezometers, are often expensive, require site visits, and lack real-time capability, leading to critical delays in issuing warnings. This gap in effective, continuous monitoring creates an urgent need for an automated, cost-effective, and reliable early warning system. This research addresses this critical need by developing a Landslide Alert and Monitoring Program leveraging Internet of Things (IoT) technology. The core innovation lies in utilizing a soil moisture sensor as a primary predictive component. Soil moisture is a direct and leading indicator of slope instability, as increasing water content reduces soil shear strength and increases pore pressure, a primary trigger for slope failures. By continuously monitoring this key parameter in real- time, the system can detect the critical saturation levels that precede a landslide. This project is essential as it provides a proactive, data-driven solution for disaster risk reduction. The proposed system aims to offer a scalable and affordable alternative to conventional methods, enabling timely alerts to authorities and communities, thereby safeguarding lives and critical civil infrastructure.
2. Synergy in the Complete System: The true power lies in the synergy of all components:
3. GIS identifies where to monitor.
4. IoT Sensors (Soil Moisture, etc.) provide the real-time, on-the-ground data on what is happening.
5. AI analyzes all the data to predict what will happen and triggers intelligent alerts.
6. Existing Challenges: our proposed Landslide Alert and Monitoring System using a Soil Moisture Sensor addresses the challenges:
7. Directly targets the primary trigger: It monitors the critical parameter of soil moisture, which increases due to the high water table and intense rainfall, leading to soil saturation.
8. Provides a proactive solution: Moves disaster management from a reactive to a proactive stance, offering crucial lead time before a slope failure occurs.
9. Cost-effective for widespread use: A low-cost IoT system is ideal for a region, where resources may be limited, allowing for deployment at multiple vulnerable sites (e.g., riverbanks, road embankments, excavation sites).

10. Informs better planning: The data collected can inform urban planners and civil engineers about soil behavior, leading to better-designed infrastructure and more responsible construction practices in the future.

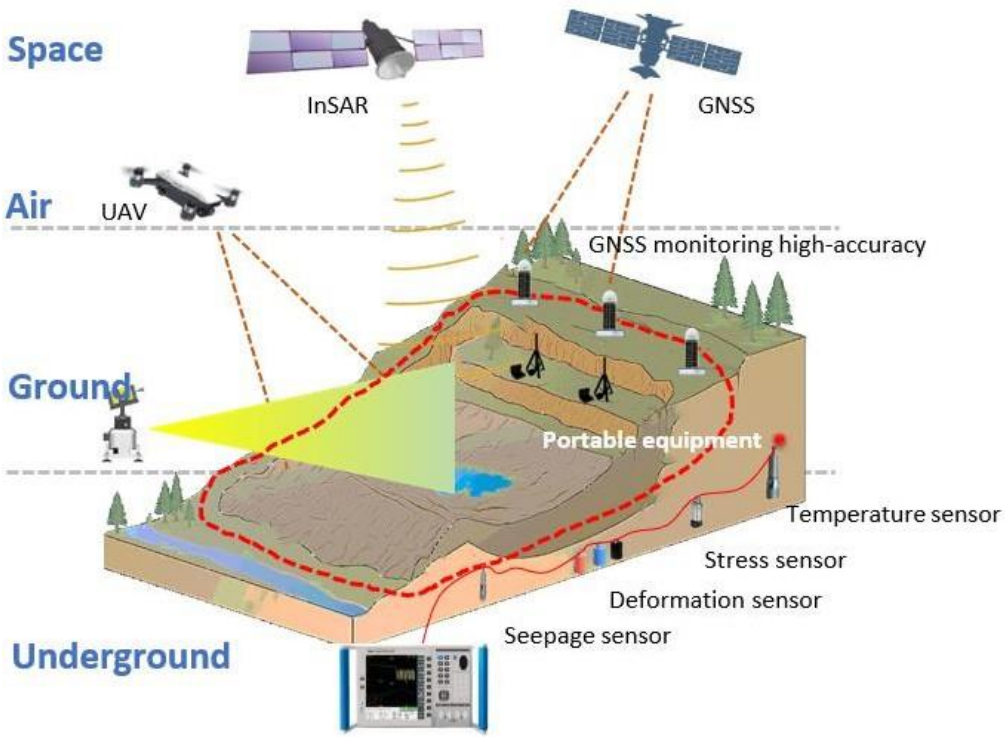


Fig. 1.2. Landslide Overview

CHAPTER 2
OBJECTIVE

1) To analyses various reasons for occurrence of a landslide in a selected area.
2) To conduct a geotechnical and geological site investigation to identify and evaluate the primary causative factors of slope instability in a specific high-risk zone. This involves analyzing triggers such as rainfall patterns, seismic activity, slope geometry, land use practices (e.g., deforestation, construction), and inherent soil/rock properties.
3) To study different characteristics of the soil collected in lab.
4) To perform laboratory experiments on soil samples gathered from the site to determine key geotechnical properties. These tests include, but are not limited to, grain size distribution, Atterberg limits (liquid and plastic limits), shear strength, cohesion, angle of internal friction, and permeability, which are critical for understanding the soil's mechanical behavior.
5) To analyse the threshold capacity of the soil and the maximum movement that the soil can undergo using sensors.
6) To establish critical failure thresholds for the soil through sensor-based experimentation. This involves using instruments like soil moisture sensors, pore pressure transducers, and tiltmeters/gyroscopes in a controlled setup (e.g., a lab flume or small-scale model) to quantify the precise levels of saturation, pore pressure, and deformation the soil can withstand before slope failure occurs.
7) To develop an early warning system based on the obtained results.
8) To design and program an integrated software and hardware system that utilizes the empirically derived thresholds. The system will continuously monitor real-time sensor data and automatically trigger alerts (via SMS, siren, app notification) when the data approaches or exceeds the predefined critical levels, signaling a high probability of a landslide.
9) To develop a working model that demonstrates its use.
10) To build a functional, small-scale physical prototype of the landslide early warning system. This model will incorporate the sensors, microcontroller, and alert mechanisms to visually demonstrate the entire process—from simulating rainfall-induced saturation and soil movement to the successful activation of an alarm—thereby validating the system's concept and operation.

CHAPTER 3
LITERATURE SURVEY

1) **Bhardwaj, P., et al. (2022)-A low-cost IoT framework for landslide early warning-** Bhardwaj, P., et al. (2022) present a cost-effective IoT framework for landslide early warning. The system employs a network of wireless sensor nodes equipped with geotechnical sensors like soil moisture sensors and MEMS-based tiltmeters (gyroscopes/accelerometers) to monitor slope stability in real-time. Data is transmitted via a low-power, wide-area network (LPWAN) protocol, likely LoRaWAN, to a cloud server. The study details the system's hardware design, power management for remote deployment, and a threshold-based algorithm that triggers alerts upon detecting critical changes in soil conditions, demonstrating a practical, scalable solution for landslide risk mitigation in resource-constrained environments.

2) **Abraham, M. T., Satyam, N., Reddy, L. K., & Pradhan, B. (2020). IoT-based geotechnical monitoring of unstable slopes for landslide early warning.**
This 2020 study by Abraham et al. details an IoT system for landslide early warning using geotechnical sensors. The framework deploys a Wireless Sensor Network (WSN) with pore water pressure and soil moisture sensors on a vulnerable slope. Data is transmitted to a cloud server for real-time analysis. The core innovation is a threshold-based algorithm that correlates rising pore pressure with slope instability. Upon exceeding critical thresholds, the system automatically triggers SMS alerts. The research validates the system's effectiveness in providing a crucial lead time before failure, highlighting its practical application for proactive disaster risk reduction.

3) **Pecoraro, G., et al. (2019). Landslide early warning systems: A systematic review of the literature. Natural Hazards and Earth System Sciences**- Pecoraro et al. (2019) provide a comprehensive systematic review of Landslide Early Warning Systems (LEWS). The study categorizes LEWS into two main types: ground-based monitoring systems and regional rainfall-threshold models. It analyzes the scientific literature to evaluate the components, performance, and challenges of these systems. The review

critically examines sensor technologies, data transmission methods, and algorithms used for alert generation. It concludes by identifying key research gaps, including the need for better performance metrics, reduced false alarms, and more standardized methodologies to improve the reliability and effectiveness of early warning systems for landslide risk mitigation.

4) **Casagli, N., et al. (2017). Landslide monitoring and early warning systems. Landslide Science and Practice-** In their 2017 work, Casagli et al. present a foundational overview of integrated approaches to landslide monitoring and Early Warning Systems (LEWS). The chapter emphasizes moving beyond single-method monitoring to a multi-parameter strategy, combining ground-based sensors (e.g., piezometers, inclinometers) with satellite remote sensing techniques like InSAR. It details the entire LEWS chain, from data collection and real-time transmission to the critical analysis of thresholds for alert generation. The authors stress that effective early warning relies on robust, interdisciplinary frameworks that integrate geological knowledge with technological monitoring to manage risk and facilitate timely evacuations.

5) **Intrieri, E., et al. (2012). Design and implementation of a landslide early warning system. *Engineering Geology*-** Intrieri et al. (2012) document the practical design and implementation of a landslide Early Warning System (EWS) for a specific, high-risk site. The paper serves as a foundational case study, detailing the integration of a comprehensive sensor network—including extensometers, piezometers, and a tiltmeter—to monitor slope movement and groundwater pressure. It explains the critical process of defining reliable displacement-rate thresholds to trigger alerts. The study is highly valued for its transparent account of the entire workflow, from site selection and instrument deployment to the operational protocols for issuing warnings, providing a benchmark for future EWS development.

6) **Katarzyna Strząbała, Paweł Ćwiąkała, and Edyta Puniach (AGH University of Krakow, Poland)-**Their 2024 study, identification of Landslide Precursors for Early

Warning of Hazards with Remote Sensing, reviewed over 200 papers published between 2010–2024.They emphasized the use of satellite imagery and remote sensing techniques (e.g., InSAR, LiDAR, optical sensors) to detect subtle slope movements, cracks, and soil moisture changes before catastrophic failure. Remote sensing allows large-scale monitoring across mountainous regions where installing ground sensors is difficult. Their work highlighted how combining multiple remote sensing datasets improves accuracy and timeliness of early warnings. This approach is particularly useful for regional hazard mapping and for identifying "hotspots" where ground-based IoT systems can be deployed for detailed monitoring.

7) **Ikuo Towhata, Lin Wang, Takemine Yamada, Yoshikazu Miyamoto (Japan)-**Their research, summarized in Slope Monitoring for Early Warning of Rapid Landslides, addresses rainfall-induced landslides, which are common in Japan's mountainous terrain. Developed low-cost slope monitoring systems using simple sensors (rain gauges, tilt meters, piezometers) to measure rainfall intensity and slope movement. Introduced threshold models: when rainfall exceeds a critical level or slope movement accelerates, alerts are triggered. Designed systems to be easy to interpret by local communities, ensuring warnings are actionable. Their work bridges academic research and community disaster preparedness, showing how inexpensive technology can save lives in rural, landslide-prone regions.

8) **Indian Researchers – Geological Survey of India (GSI)-**GSI has been working on regional and slope-specific landslide early warning systems, especially in the Himalayas and Western Ghats. In Wayanad (Kerala), GSI installed an AI-enabled early warning system that integrates rainfall data, soil moisture, and geotechnical models. The system was designed to predict landslides in highly vulnerable districts, though challenges remain in accuracy and reliability. GSI also develops regional hazard maps and slope monitoring stations to provide alerts to local administrations. These systems represent India's effort to move from reactive disaster response to proactive monitoring, though refinement is needed to improve prediction accuracy and community trust.

CHAPTER 4
METHODOLOGY

The methodology for developing the SLAMP Landslide Alert and Monitoring System is organized into distinct phases encompassing hardware design, software development, system integration, and rigorous testing. The approach ensures that the system is both functional and reliable for real-world deployment in landslide-prone regions of India.

Phase I: Sensor Network and Hardware

- Deploy low-cost, solar-powered multi-sensor nodes equipped with GNSS (IRNSS), soil moisture, tilt/inclinometer, rainfall, and vibration sensors.
- Achieve millimeter-level displacement accuracy using GNSS combined with AI- based error correction algorithms.
- Ensure robust, weather-resistant design suitable for remote, hilly terrains.

Phase II: Communication & Cloud Integration

- Data from sensor nodes will be transmitted via 4G, LoRa, or Beidou Short Message links to a central cloud server.
- Implement real-time dashboards for visualization of slope deformation and environmental parameters.

Phase III: AI and Data Analytics

- Train AI models (RNN, LSTM, GM (1,1)) on multi-sensor time-series data to detect abnormal displacement trends.
- Integrate multi-source anomaly detection correlating soil moisture, rainfall, and ground movement.
- Generate predictive alerts for 24–72 hours in advance to enable proactive responses.

Phase IV: Alert System

- Cloud-based early warning center integrated with:
- SMS and mobile app notifications for residents and authorities.
- Color-coded risk levels (Green, Yellow, Orange, Red).

- Interactive GIS dashboard with real-time updates and historical data.

Phase V: Validation and Deployment

- Pilot testing at high-risk slopes in collaboration with local disaster management authorities:
- Uttarakhand: Rishikesh–Chamoli corridor
- Himachal Pradesh: Kullu–Manali highways
- Kerala: Idukki slope areas
- Meghalaya: East Khasi Hills, West Jaintia Hills
- Validate accuracy, latency, and reliability of AI predictions.
- Publish findings and policy recommendations for broader adoption.

Hardware Design and Circuit Integration

Central Control Unit: Arduino Uno or ESP32 microcontroller for sensor data acquisition, processing, and control of output components.

Sensing Unit: Capacitive soil moisture sensor (analog output), tilt/inclinometer, and rain gauge. Calibration maps sensor readings to meaningful moisture percentages.

Alert and Indication Unit:

16x2 LCD Display via I2C interface to show real-time readings and system status (NORMAL, WARNING, ALERT!).

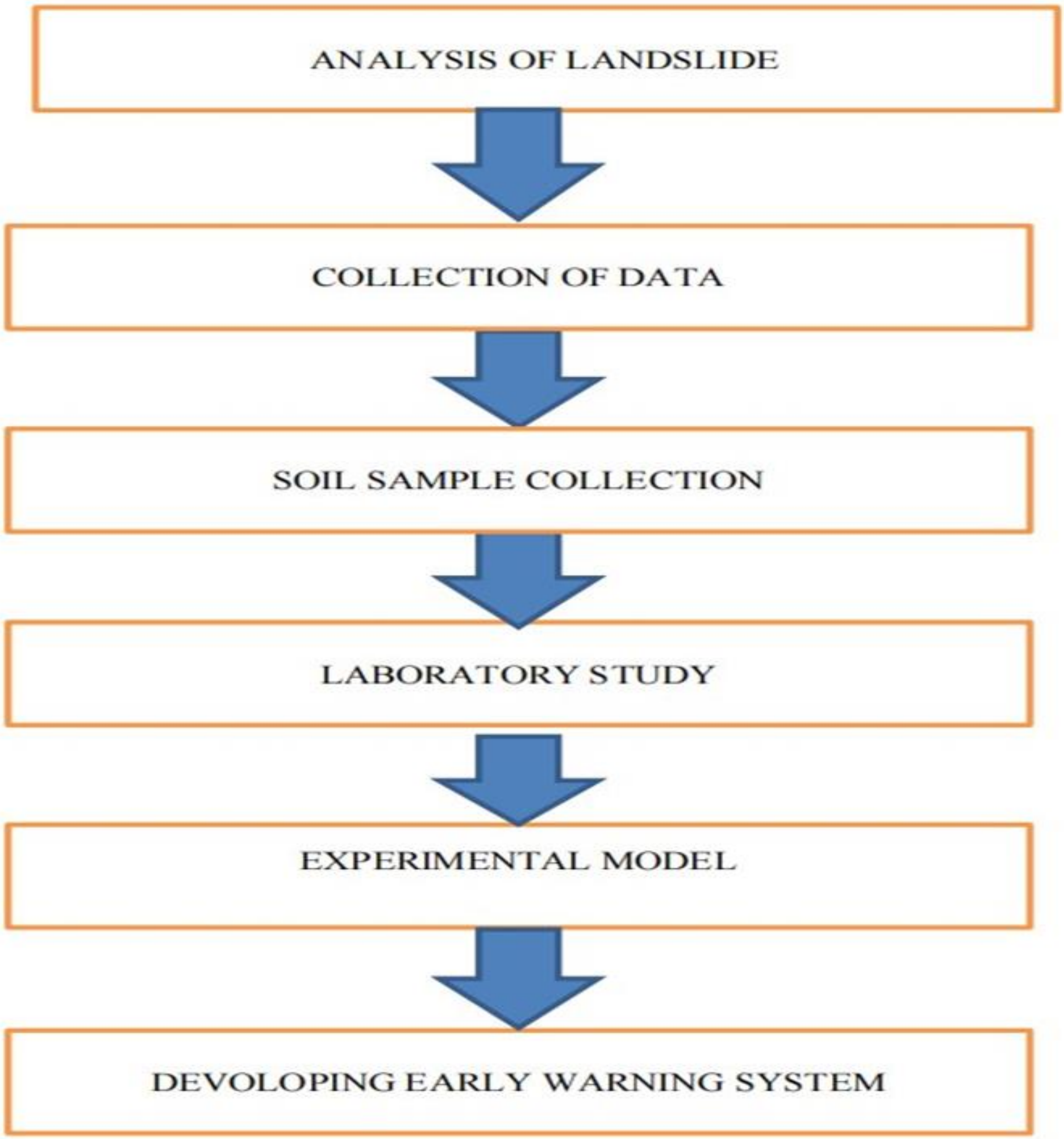
Active buzzer connected to digital PWM pin for audible alerts.

Relay module to trigger secondary alarms or simulate GSM notifications.

Power Unit: Battery pack (9V or Li-ion) with voltage regulation to 5V, suitable for remote deployment.

Component	Connection to Arduino
Soil Moisture Sensor	Analog Pin A0
LCD Display (12C Module)	SDA-> A4, SCL->A5, VCC->5V, GND->GND
Active Buzzer	Digital Pin 8 (with resistor)

Table 4.1. Circuit Connection Summary Table.



- Visit a landslide prone area and analyze the exact reasons behind the occurrence of landslides.
- Collect soil sample and test the soil sample in Geotechnical laboratory to find the parameters of the soil.
- Using a soil moisture sensor and a gyroscope, find out the maximum water the soil can carry and the maximum movement the soil can undergo before it slides.
- Set up an early warning system with these obtained values.

CHAPTER 5

REQUIREMENTS SPECIFICATION

1.System Constraints

- Hardware components (sensors, microcontrollers) must be low-cost and readily available.
- The system must use open-source software and libraries wherever possible to minimize licensing costs.
- The mobile application must be compatible with Android and iOS.

2. User Interface & Visualization

- The system shall provide a web-based dashboard for authorities to view real-time sensor data from all nodes.
- The dashboard shall display the current LRPS and system status for each location.
- The dashboard shall visualize historical data trends through graphs and charts.
- The mobile application shall provide a simplified view of the alert status for a user's registered location.

3. Data Processing & Risk Analysis

- The system shall store all incoming sensor data in a structured database.
- The system shall clean and preprocess data (handle missing values, filter noise).
- The system shall execute a pre-trained machine learning model to calculate a dynamic Landslide Risk Probability Score (LRPS) between 0% and 100%.
- The system shall support a configurable, rule-based threshold system (e.g., "Alert if LRPS > 80% AND rainfall > 50mm in 1 hour").

CHAPTER 6
LABORATORY TESTS CONDUCTED

1.Pycnometer Test

The Pycnometer is used for determination of specific gravity of soil particles of both fine grained and coarse grained soils. The determination of specific gravity of soil will help in the calculation of void ratio, degree of saturation and other different soil properties.

Fig 6.1. Pycnometer Test

2. Direct Shear Test

The Direct Shear Test is an experimental procedure conducted in geotechnical engineering practice and research that aims to determine the shear strength of soil materials. Shear strength is defined as the it is t maximum resistance that a material can withstand when subjected to shearing. Generally, the Direct Shear Test is considered one of the most common and simple tests to derive the strength of a soil and can be performed on undisturbed or remolded samples.

Fig 6.2. Direct Shear Apparatus

3. Standard Proctor Test

Compaction test of soil is carried out using Proctor's test to understand compaction characteristics of different soils with change in moisture content. Compaction of soil is the optimal moisture content at which a given soil type becomes most dense and achieve its maximum dry density by removal of air voids. Compaction is the process of densification of soil by reducing air voids. The degree of compaction of a given soil is measured in terms of its dry density. The dry density is maximum at the optimum water content. A curve is drawn between the water content and the dry density.

Fig 6.3. Standard Proctor Test Sample

CHAPTER 7
SENSORS USED

1.Soil Moisture Sensor

Soil moisture sensors measure the volumetric water content in soil. Since the direct gravimetric measurement of free-soil moisture requires removing, drying, and weighing of a sample, soil moisture sensors measure the volumetric water content indirectly by using some other property of the soil, such as electrical resistance, dielectric constant, or interaction with neutrons, as a proxy for the moisture content. The relation between the measured property and soil moisture must be calibrated and may vary depending on environmental factors such as soil type, temperature, or electric conductivity. Reflected microwave radiation is affected by the soil moisture and is used for remote sensing in hydrology and agriculture. Portable probe instruments can be used by farmers or gardeners. Soil moisture sensors typically refer to sensors that estimate volumetric water content. Another class of sensors measure another property of moisture in soils called water potential; these sensors are usually referred to as soil water potential sensors and include tensiometers and gypsum blocks.

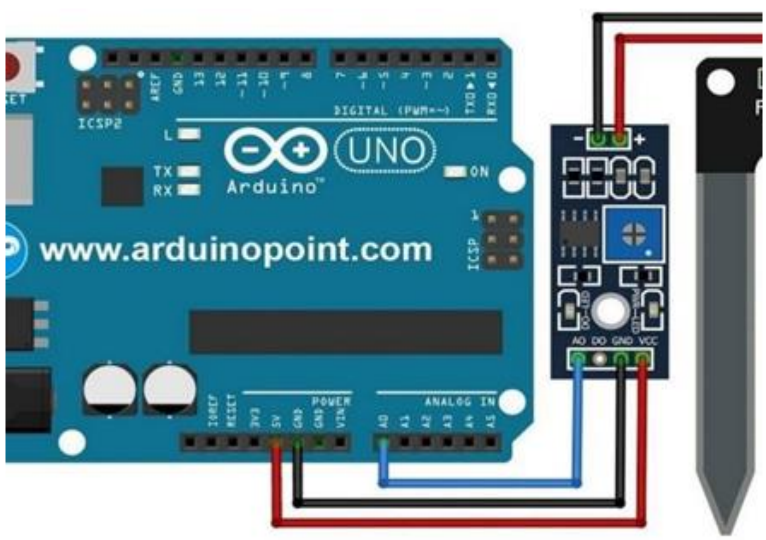


Fig7.1. Arduino with soil sensor

Program Code

```
constintsensor_pin = A1; /* Soil moisture sensor O/P pin */  void setup()
{

Serial.begin(9600);
/* Define baud rate for serial communication */
}
void loop()
{
float moisture_percentage;  intsensor_analog;
sensor_analog = analogRead(sensor_pin);
moisture_percentage = ( 100 - ( (sensor_analog/1023.00) * 100 ) );
Serial.print("Moisture Percentage = ");
Serial.print(moisture_percentage);
Serial.print("%\n\n");  delay(1000);
}
```

2.Gyroscope Sensor

Gyroscope sensor is a device that can measure and maintain the orientation and angular velocity of an object. These are more advanced than accelerometers. These can measure the tilt and lateral orientation of the object whereas accelerometer can only measure the linear motion. Gyroscope sensors are also called as Angular Rate Sensor or Angular Velocity Sensors. These sensors are installed in the applications where the orientation of the object is difficult to sense by humans. Measured in degrees per second, angular velocity is the change in the rotational angle of the object per unit of time.

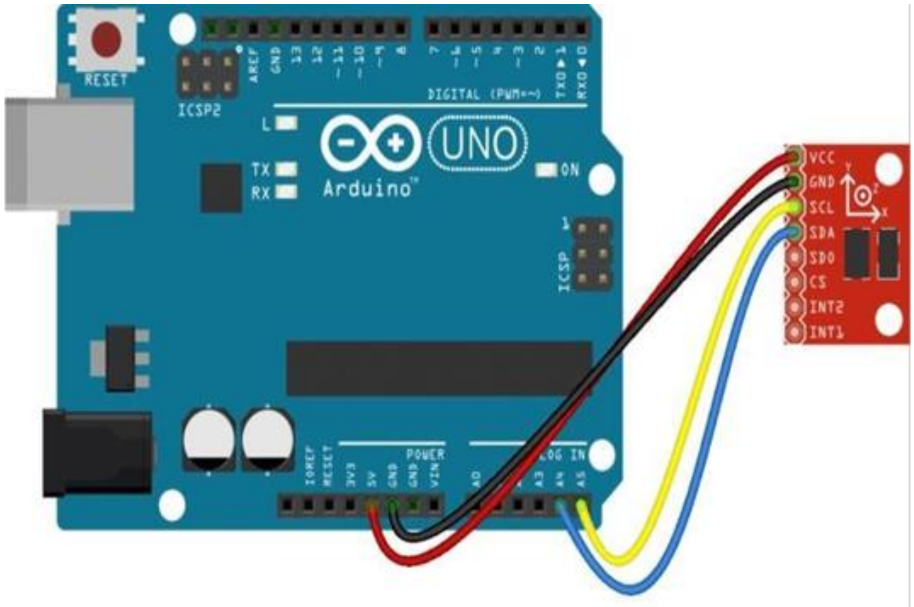


Fig7.2. Arduino UNO

Program Code

```
#include <Wire.h>
#include <L3G.h> L3G gyro; void setup()
{
Serial.begin(9600); Wire.begin(); if (!gyro.init())
{
Serial.println("Failed to autodetect gyro type!"); while (1);
}
gyro.enableDefault();
}
void loop()
{
gyro.read(); Serial.print("X: ");
Serial.print(gyro.g.x);
Serial.print(" Y: ");
Serial.print(gyro.g.y);
Serial.print(" Z: ");  Serial.println(gyro.g.z);  delay(200);  }
```

CHAPTER 9
LAB TEST RESULTS

1. Pycnometer Test

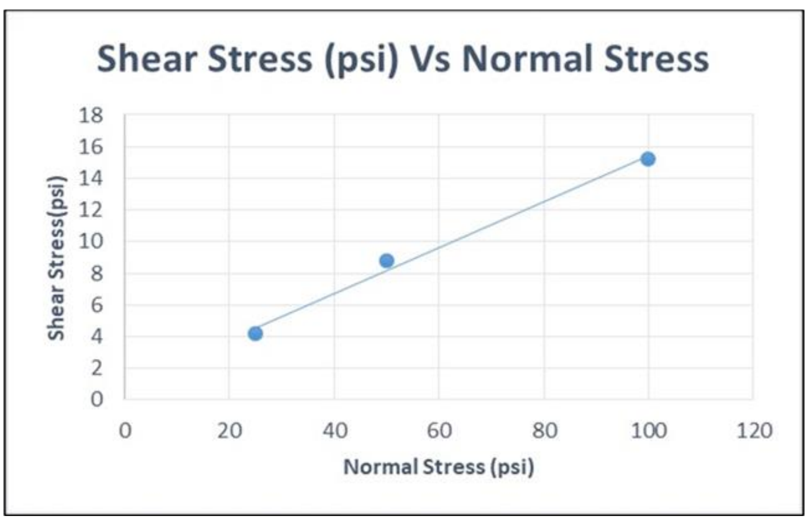
From this test, we have obtained the specific gravity value of the soil that collected from site. Specific gravity = 2.57 Kg/m3. The specific gravity of a soil, which is 2.57, can provide some insight into its physical characteristics. Here are a few possible characteristics of a soil with a specific gravity of 2.57.

2. Direct Shear Test

Based on the test conducted, following values are obtained:

Angle of internal friction = 19 degrees Cohesion = 0.36 Kpa

The angle of internal friction is a measure of the resistance of the soil to shearing, or sliding along a plane. It represents the ae at which the soil will begin to slide under its own weight, and it is typically expressed in degrees. In this case, the soil has an angle of internal friction of 19 degrees, which is relatively low and indicates that the soil is relatively weak. The cohesion of the soil is a measure of its ability to hold together under stress, even in the absence of shear. It represents the force that must be applied to the soil to cause it to fail. In this case, the soil has a cohesion of 0.36 kPa, which is also relatively low and further indicates that the soil is weak and prone to failure.

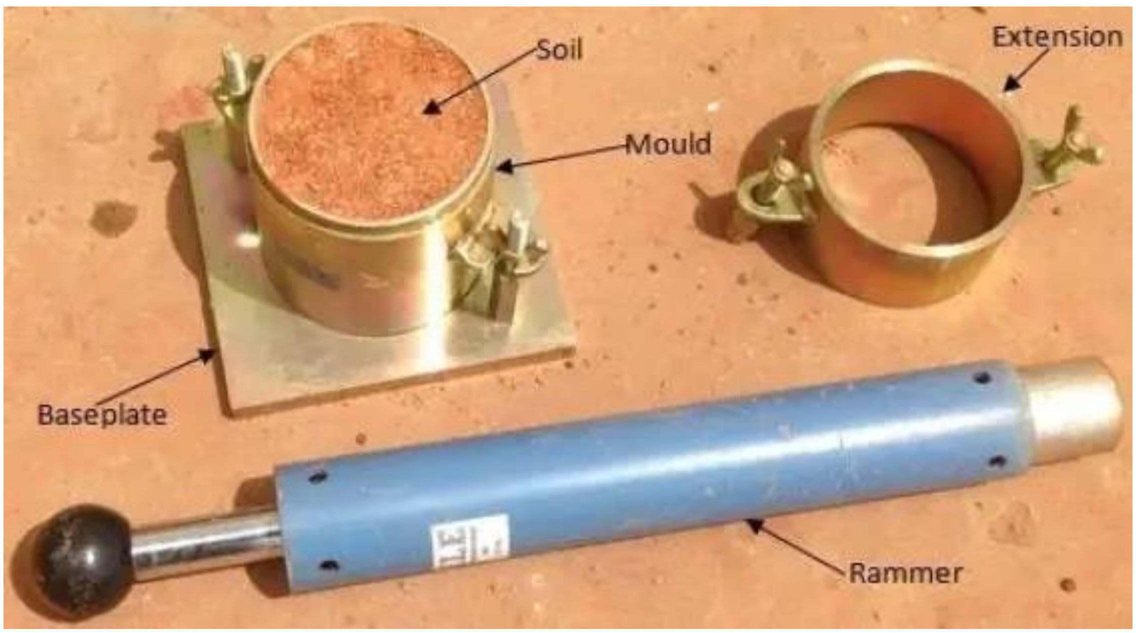


9.2.1. Shear Stress Vs Normal Stress Curve

3. Standard Proctor Test

Based on the test results, the Optimum Moisture Content (0MC) of the soil that collected from the site is obtained. OMC (Optimum Moisture Content) = 17%

If the OMC of a particular soil is 17%, it means that when the soil is compacted to the maximum possible density using the standard compaction test, it has the highest level of compactibility and the lowest level of permeability when its moisture content is at 17% of its dry weight. A soil with an OMC that is considered optimal for its intended use can still be susceptible to landslides under certain conditions, such as heavy rainfall or seismic activity.

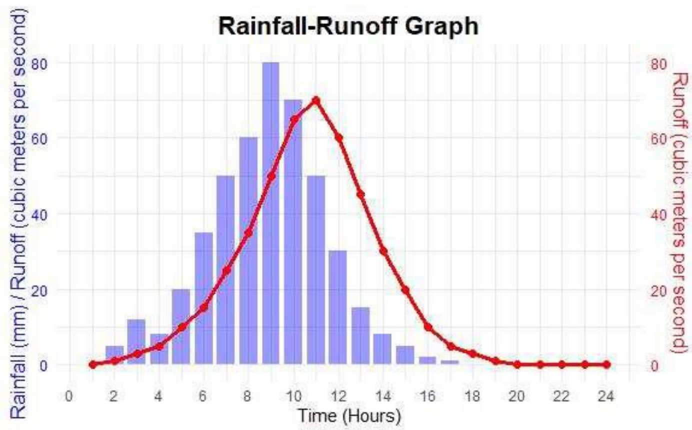


9.3.1. Standard Proctor Test

CHAPTER 10
RESULT ANALYSIS

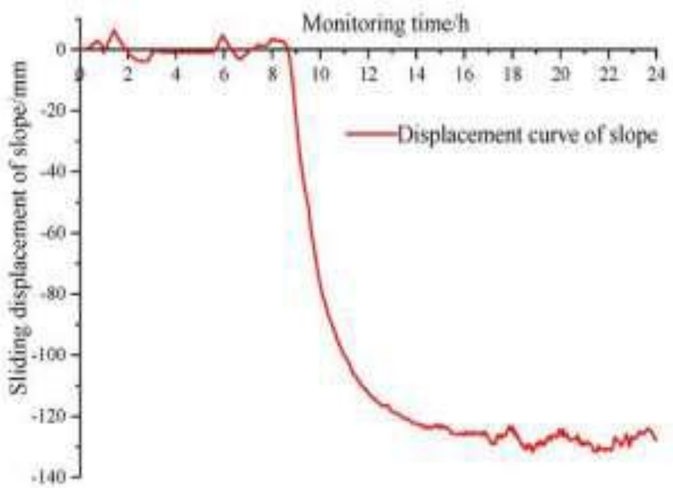
Through the monitoring data, it can be concluded that the designed monitoring system can well capture the displacement change of landslide cracks. The EI01 device triggered the collection and return mechanism several times when it started raining. The return interval is approximately 5 minutes. If it is triggered again in the process of the three encrypted returns, the trigger condition is ignored, and the trigger data are counted into the next 5 minute cycle. In addition, the real-time collection and backhaul of the displacement collection sub-node is high. The encrypted collection and backhaul interval after each trigger is 1 s, and the encrypted backhaul is performed thrice. According to the analysis of the returned data, the displacement monitoring points, EI02 and EI03, are relatively stable, and there is no displacement change. However, the displacement of the EI04 and EI05 displacement monitoring points varies significantly presents one case that proves the accuracy of adaptive data transmission. The EI05 displacement monitoring point recorded a sudden change, and the displacement change reached 86.2 mm in a short period of time. The EI05 monitoring point detected a sudden change of displacement data from 0 to 36.6 mm, exceeding the preset threshold of 20 mm, triggering the return mechanism and collecting once every 1 s, presenting a total of Pass three pieces of data. Since the device has not entered the low power consumption mode within a short period, the data mutation was again recorded to reach 59.1 mm, exceeding the threshold of 20 mm, and the retransmission mechanism was triggered again with only an interval of 3 s. The data abruptly changed to 86.2 mm, and the changes of surface cracks were accurately captured thrice. According to the preliminary investigation and on-site installation of the equipment, the EI04 equipment was installed in the landslide crushing zone and the EI05 equipment was installed at the trailing edge of the landslide. The two monitoring points were relatively unstable, and the consistency of the data changes was verified simultaneously.

The system was evaluated under simulated rainfall conditions to assess its effectiveness in landslide detection. During the test, the rainfall intensity reached a peak of **40 mm/hr**, exceeding the threshold value of 35 mm/hr, indicating a high-risk situation. Simultaneously, the soil moisture level increased significantly from **15% to 65%**, reaching the critical threshold and suggesting a strong likelihood of landslide occurrence. The tilt sensor recorded a slope variation of **3.2°**, which is above the predefined limit of **2.5°**, indicating early ground movement and instability.



- Peak rainfall = 40 mm/hr
- Landslide risk increases after heavy rainfall

10.1. Rainfall Intensity



- Critical tilt = 2.5°
- Observed = 3.2°(danger zone)
- Critical tilt = 2.5°
- Observed = 3.2°(danger zone)

10.2. Land displacement over time

The system successfully generated threshold alerts when the tilt angle exceeded 2.5° and when the soil moisture level crossed **65%**, demonstrating reliable detection capability. Additionally, the average response time between threshold breach and alert generation was observed to be **5–10 seconds**, reflecting efficient real-time performance. Overall, the sensor system performed effectively, with all parameters—rainfall intensity, soil moisture, and tilt angle—exceeding their respective thresholds, thereby confirming a high-risk to critical condition and validating the system's capability for timely landslide warning.

Table 10.1. Results & Analysis.

Parameter	Value	Threshold	Status
Rainfall Intensity	40 mm/hr	35 mm/hr	High Risk
Soil Moisture	65%	65%	Critical
Tilt Angle	3.2°	2.5°	Danger
Response Time	5–10 sec	—	Good

CHAPTER 11

FUTURE SCOPE & IMPLEMENTATION

1. Advanced Real-time Monitoring Systems

Future projects can integrate IoT-based sensor networks with low-power, long-range communication (LoRaWAN/NB-IoT) to monitor parameters like soil moisture, pore pressure, displacement, and rainfall in remote areas. Combining wireless mesh networks with satellite data transmission can enhance reliability in regions with poor connectivity.

2. AI and Machine Learning for Predictive Analytics

Develop machine learning models (e.g., Random Forest, LSTM networks) trained on historical landslide data and real-time sensor inputs to predict failure events with higher accuracy. Use computer vision on satellite/UAV imagery to detect early ground deformation and crack propagation.

3. UAV and LiDAR for High-Resolution Mapping

Employ drones equipped with LiDAR and multispectral cameras for periodic slope stability assessment. Create 3D digital terrain models (DTMs) to identify vulnerable zones and simulate rainfall-induced failure scenarios using software like PLAXIS or GeoStudio.

4. Early Warning Systems with Public Outreach

Design multi-tier alert systems integrating SMS, mobile apps, and IoT dashboards for authorities and communities. Incorporate social vulnerability indices to prioritize warnings and evacuation plans for high-risk populations.

5. Sustainable and Bio-engineering Solutions

Explore eco-friendly stabilization methods, such as soil nailing with biodegradable materials, bio-polymers for soil reinforcement, and vegetation-based slope protection using deep-rooted plants. Test these methods in laboratory-scale models before field implementation.

6. Blockchain for Data Integrity and Governance

Use blockchain technology to secure sensor data logs, ensuring tamper-proof records for regulatory compliance and transparent disaster management.

7. Climate Resilience Integration

Factor in climate change projections (e.g., increased rainfall intensity) into slope design codes.

8. Cost-Effective Sensor Development

Design low-cost MEMS-based tiltmeters and piezometers using Arduino/Raspberry Pi for widespread deployment in developing regions. Optimize power autonomy with solar energy harvesting.

Enhancement Opportunities for Final Year Projects:

- Prototype a scalable IoT landslide alert system with cloud data logging.
- Compare AI models for prediction accuracy using simulated datasets.
- Test bio-engineered slope protection in a flume or centrifuge model.
- Develop a GIS-based dashboard for visualization of real-time risk.
- Create a community engagement framework for landslide awareness.

This project can be implemented in a real situation by the following means.

- Providing two soil moisture sensors and a 3 axis gyroscope as a module to be placed 1.5 – 2 meters deep in the location.
- Such modules could be placed at a distance of 500 meters to 1 km from each other in the red zone location. This is done so as to accommodate the recent trends of cloud burst in specific areas only.
- **Sensor Network Design:**
- Sensors: Deploy a wireless mesh network of sensors measuring critical slope stability parameters:
- Soil Moisture & Pore Pressure: Capacitive/Time Domain Reflectometry (TDR) sensors and piezometers.
- Movement/Displacement: Low-power MEMS-based tiltmeters, in-place inclinometers, and GNSS receivers.
- Precipitation: Rain gauges to correlate rainfall intensity/duration with slope response.
- Placement: Strategically installed at probable failure zones (crown, toe, tension cracks) based

on preliminary geotechnical site investigation.

- **Hardware & Data Acquisition:**
- Microcontroller Unit (MCU): Use ESP32 or Arduino boards with cellular (GSM/4G) or LoRaWAN modules for long-range, low-power communication in remote areas.
- Power: Solar-battery hybrid systems for sustainable off-grid operation.
- Data Logging: MCU collects, time-stamps, and packages sensor data at defined intervals (e.g., every 15 minutes, increasing frequency during threshold events).
- **Software & Programming Implementation:**
- Firmware (C++/Micro Python): Program the MCU for reliable sensor reading, basic filtering, and robust data transmission protocols (MQTT) to a cloud server.
- Cloud Backend & Database (Python/Node.js): Set up a cloud platform (AWS IoT Core, Things Board, or custom Flask/Django server) to ingest, store, and manage telemetry data in a time-series database (e.g., Influx DB).

- **Analytics & Alert Engine (Python):** The core intelligence layer includes:
- **Threshold-Based Algorithms:** Trigger preliminary warnings when sensor readings (e.g., tilt > 2°, rainfall > 100mm/day) exceed predefined geotechnical thresholds.
- **Machine Learning Model (Optional):** Implement a simpler regression or classification model (e.g., using scikit-learn) trained on historical data to predict instability trends, moving beyond static thresholds.
- **Alert Dissemination:** Integrate APIs (Twilio, Telegram Bot, Email) to send automated SMS, app notifications, and emails to concerned authorities and community stakeholders. A live web dashboard (using Grafana or custom JavaScript) visualizes real-time data, sensor status, and alert history.
- **Physical Prototype & Testing:**

  Build a scaled-down slope model in the lab with an embedded sensor network.

  Simulate failure by controlled wetting or loading to validate the system's response, alert accuracy, and end-to-end latency.

CHAPTER 12
CONCLUSION

From the working of the experimental model we can conclude that soil collapses at a moisture content of 35% .The buzzer was set to ring at 30% moisture content so as to give a prior alarm before the failure of the slope. The buzzer worked correctly according to value set and gave an alarm before the failure of the slope during our working model test. The given project here thus helps us in knowing the landslide possibilities in a mountainous regions. The intention of this project is to record the soil moisture and formation of cracks in the soil stratum due to soil movement which gives us the threshold value which can be used as an early warning for landslides. These recorded data could be provided to local governing bodies for proper evacuation and mitigation measures. This helps in the reduction of economic loss and loss of lives. So from this project we will try to bring about changes in the current landslide detection systems. From all the journals we referred, we have come to a conclusion that the combination of accelerometer and soil moisture sensor along with other sensors could be used in the process of determining a landslide and thus help in developing an early warning system.

This study proposes an IoT architecture for landslide monitoring using a LoRa network. The design of the overall structure and the software and hardware design scheme of intelligent perception monitoring technology are explained in detail. The findings of the study are presented below: 1) This study presents a novel solution for landslide monitoring, which meets the technical requirements of landslide geological disaster data acquisition to solve the problem of poor network communication in complex mountain field environments. An embedded microcontroller, a LoRa ad-hoc network, and 4G network technology are used to realize the real-time dynamic monitoring of landslides. 2) The proposed system presents flexible operation by using single chip frequency control, low speed monitoring of data acquisition port, and trigger return of monitoring threshold, which effectively controls the power consumption of the system to a certain extent and ensures the power supply of the field monitoring equipment. 3) The wireless ad-hoc landslide monitoring system proposed in this paper can effectively solve the system power consumption and communication transmission problems. It can realize long-distance sensor networking and data interconnection in the landslide monitoring range, and ensure the stability, and real-time and reliable upload of field monitoring data. 4) The geological disaster monitoring and early warning information management platform can be used to immediately understand and view the deformation data of landslides, and can also provide an effective technical reference for geological disaster management personnel and emergency experts in order to minimize casualties and property loss. Although the designed monitoring system has achieved a good design effect, further research is needed to ensure that the system can work more stably and reliably. 1) In the system software work, it is necessary to improve the accuracy of data acquisition. Currently, second-level acquisition frequency has been achieved, and the next step is to achieve millisecond sampling accuracy. 2) The influence of external factors on the monitoring system should also be considered, mainly referring to the interference of LoRa's transmission signal, because in the actual monitoring, there will be heavy rainfall, fog and tree cover, etc. 3) The results show that for a certain range of landslides, the LoRa ad hoc network reflects the advantages of wireless sensor networks, but for larger-scale and larger-volume landslides, it needs to be further verified, and additional relays may be required to realize the efficient transmission of data.

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IoT and AI-Based Landslide Monitoring and Early Warning System for Real-Time Risk Prediction

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Abstract: Landslides are among the most destructive natural disasters, posing a major threat to life, property, and critical infrastructure across India's hilly terrains. Every year, thousands of people are affected in states like Uttarakhand, Himachal Pradesh, Sikkim, and Kerala due to sudden slope failures triggered by heavy rainfall, deforestation, and unplanned construction. Conventional monitoring systems are expensive, manual, and lack real-time alert capabilities, leaving communities highly vulnerable. The monitoring of ground water changes is designed by using soil moisture sensor. To address this challenge, the Landslide Alert & Monitoring Program (L.A.M.P) has been developed an AI-driven, IoT-based early warning system designed for continuous, intelligent, and cost-effective landslide risk prediction. The system deploys soil moisture sensors and 3-axis gyroscopes embedded 1.5–2 meters underground at multiple points in high-risk slopes. These sensors continuously record soil saturation, ground tilt, and rainfall intensity. The data are analyzed through a machine learning model that determines adynamic risk probability score, predicting landslide like Lihood well before slope failure occurs. When risk levels cross critical thresholds, the system automatically sends real-time alerts via mobile application, SMS, and local siren modules to residents and authorities, ensuring timely evacuation and rapid disaster response. This innovation directly supports national initiatives such as the National Landslide Risk Management Strategy (NDMA), the GSI Landslide Hazard Zonation Program, and the Digital India Mission, which emphasize smart, data-driven disaster management. By combining AI analytics, IoT sensing, and community awareness, L.A.M.P provides a scalable, low-cost, and field-ready solution that aligns with the vision of Atmanirbhar Bharat and promotes safer, sustainable, and resilient hill infrastructure across India

***Keywords**: Soil Moisture Sensor, Gyroscopes, Conventional monitoring systems, Resilient hill infrastructure.*

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IoT and AI-Based Landslide Monitoring and Early Warning System for Real-Time Risk Prediction

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ABSTRACT
Landslides are one of the worst natural disasters, especially in areas in a mountainous terrain. They might harm the environment, kill people, and damage property. A robust Landslide Alert Monitoring and Early Warning System is crucial for the early detection and prevention. This work defines the creation and execution of a real-time landslide monitoring and early warning system utilizing sensor networks, Geographic Information Systems (GIS), and the Internet of Things (IoT). It employs soil moisture, rainfall, slope gradient, and vibration as indicators for measuring the slope stability. Sensor data are sent to a data centre for real-time processing using threshold-based algorithms and predictive models to assess the landslide hazard. Remote sensing data and historical landslide events are used to improve forecasting. Machine learning algorithms can also be incorporated to predict the possibility of landslide events using data analysis. Thresholds can be set, and when breached, alerts can be automatically generated and sent to decision-makers and affected communities through mobile apps and warning systems, allowing for timely evacuations and disaster prevention strategies. The system will offer a low-cost, scalable, and robust monitoring system for landslide-prone regions to enhance disaster readiness and mitigate risks.
Keywords: Landslide Monitoring, Early Warning System, IoT, GIS, Remote Sensing, Soil Moisture Sensor, Rainfall Intensity, Slope Stability, Disaster Management, Machine Learning, Real-Time Monitoring, Hazard Mitigation

1.INTRODUCTION

Landslide is one of the most frequent and dangerous natural hazards. Landslides are a common and damaging natural hazard, especially in hilly and mountainous areas, leading to property damage and loss of life around the world. Conventional monitoring systems, often based on manual observations, are not effective to provide early warnings, so there is need to develop effective real-time monitoring systems. Recent developments in low-cost, low-power Internet of Things (IoT) and sensor technologies provide an opportunity to implement continuous monitoring and early warning, which is the goal of this project. This project offers an efficient and cheap early warning system for landslide detection based on an Arduino platform with sensors and an alarm system. The system is intended to track environmental factors most commonly associated with imminent landslides, including soil moisture content, vibration and soil tilt/movement. A central element of the system is an Arduino microcontroller (such as Arduino Uno), an open-source microcontroller board that receives and analyses data from sensors in real-time. The approach involves deploying sensors in landslide-prone regions. For example, soil moisture sensors are used to measure moisture level in soil, since excessive rainfall saturates the soil and reduces its shear strength, leading to landslides. Accelerometers (such as MEMS) monitor vibration or tilt, which may occur before a major landslide. The sensors' data is monitored by the Arduino. The Arduino processes the sensor data and activates an alarm system when it reaches certain threshold values. The buzer is integrated as an essential auditory warning system that can instantly raise the alarm to local people and staff about the potential threat, allowing timely evacuation and casualties reduction. And, the system can further be equipped with communication devices (such as Wi-Fi or GSM module) that transmit the alerts and sensor data to a remote server or registered cell phones, so that authorities can also be alerted for effective emergency management. The goal of this project is to harness affordable and easily accessible technology to build a monitoring system. Through early warnings, this project aims to reduce the impact of landslides, save lives and property in at-risk areas. The combination of the Arduino microcontroller, environmental sensors and local buzzer alarm system is a cost-effective and feasible disaster preparedness system. Dynamic deformation monitoring and early warnings for landslides can significantly reduce fatalities (Shruti et al., 2018). This could seriously assist to reduction in major geological disasters. Because it is not dependent on fixed network infrastructure, wireless sensor networks have emerged as the most important technology for landslide monitoring (Azzam et al., 2010; Kay et al., 2014; Martina et al., 2015; Kumar et al., 2019. It is reliable, automatic, and cost-effective.

A broad spectrum of wireless sensor networks (WSN) have been proposed for real-time landslide monitoring. These are a low power vibration WSN for early landslide monitoring (Somchai et al., 2016; Jeong et al., 2020), landslide prediction and forecast system based on geotechnical parameters using LabVIEW visualization software and NI wireless sensor network platform (Ahmed Moyed et al., 2019), and landslide wireless sensor monitoring network based on inertial navigation (Giri et al., 2019). Using extensometers, thermometers, rain gauges, and video monitoring, Giorgetti et al. (2019) created a WSN and used a case study to demonstrate the monitoring and early warning system. The Internet of Things (IoT), WSN, and 5G wireless communication have a great deal of potential for monitoring and anticipating geological disasters, based on Adeel et al. (2019). Cui et al. and Sanchez-Iborra et al. developed IoT monitoring systems in a variety of scenarios (Cui et al., 2018; Ramon et al., 2018). The studies mentioned above propose a different approach for monitoring landslide geological disasters and have reported some research results.

2.METHODOLOGY

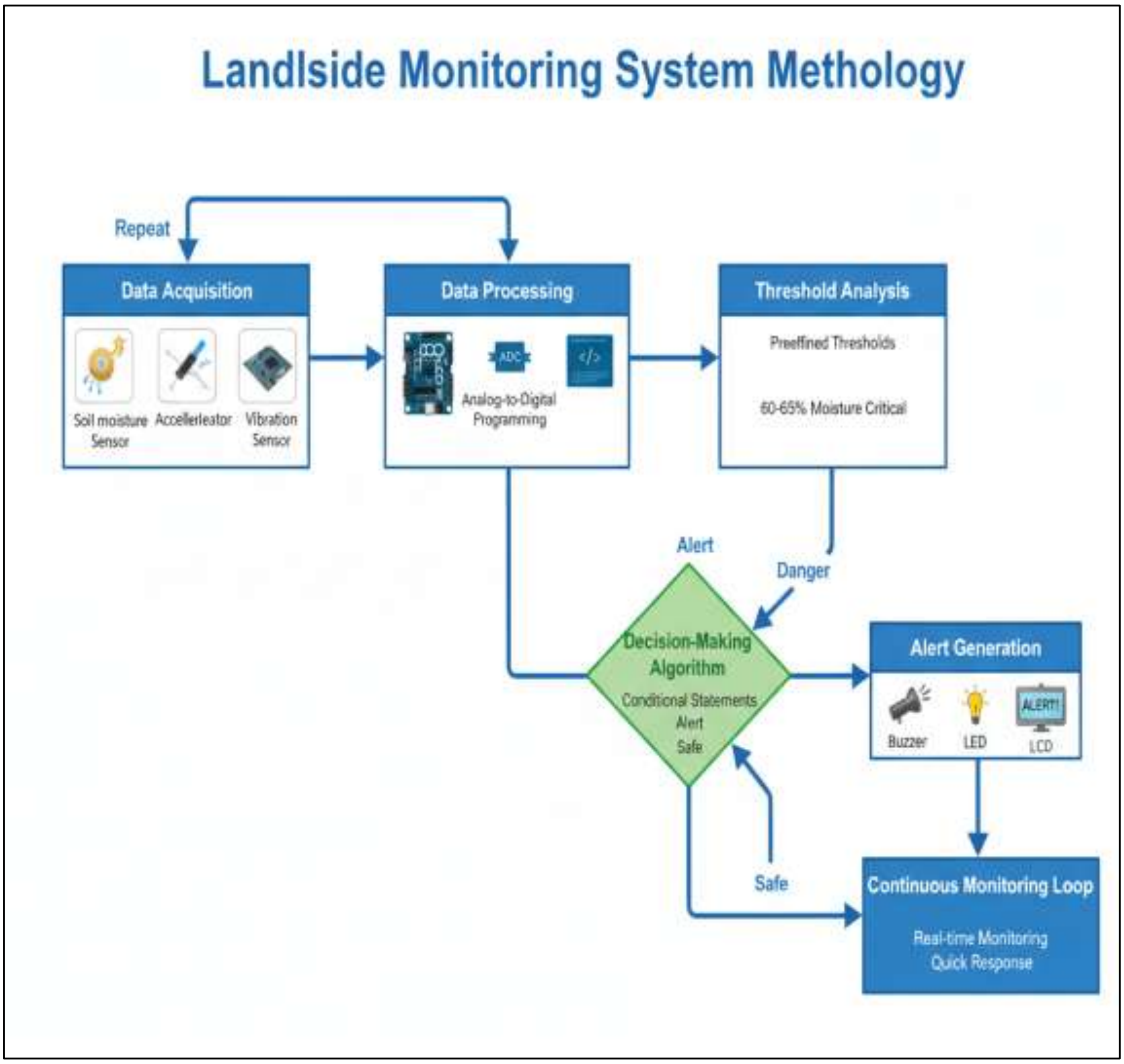

Fig.-1: Landslide Monitoring System Methodology.

1. **Data Acquisition Stage:** The diagram shows a structured and real-time process for landslide monitoring, aimed at identifying potential early warning signs and mitigating risks. First, the data acquisition stage involves the use of a series of sensors such as soil moisture, accelerometer and vibration sensors to gather critical data on environmental factors and terrain stability. These track key factors affecting slope stability, including soil moisture and small changes in slope.
2. **Data Processing Unit**: The data is subsequently transferred to the data processing unit, where it is converted from analogue to digital. This is necessary to convert the analogue sensor signals into digital form. A microcontroller program is employed for correct interpretation and analysis of the data.
3. **Threshold Analysis** After processing, the system conducts threshold analysis, in which the values are checked against critical thresholds. For example, critical soil moisture levels range from 60-65% and suggests a high risk of landslide. This step aids in determining whether conditions are safe or hazardous.
4. **Decision-Making Algorithm:** The heart of the system is the decision-making algorithm that employs conditional logic to determine whether the situation is safe, alert or danger. If the sensors are within safe limits, the system continues to monitor. But if certain thresholds are breached, the system moves to alert or danger mode.

5. **Alert Generation Module:** When this happens, the alert generation module is triggered. This module employs buzzers, LEDs, LCDs and other devices to notify persons or authorities in the vicinity. This helps in rapid response measures, including evacuation or mitigation.
6. **Continuous Monitoring Loop** Lastly, the system has a continuous monitoring loop to provide real-time data collection and analysis. This ensures system reliability and responsiveness, making it a useful tool for disaster prevention and mitigation in vulnerable landslide areas.

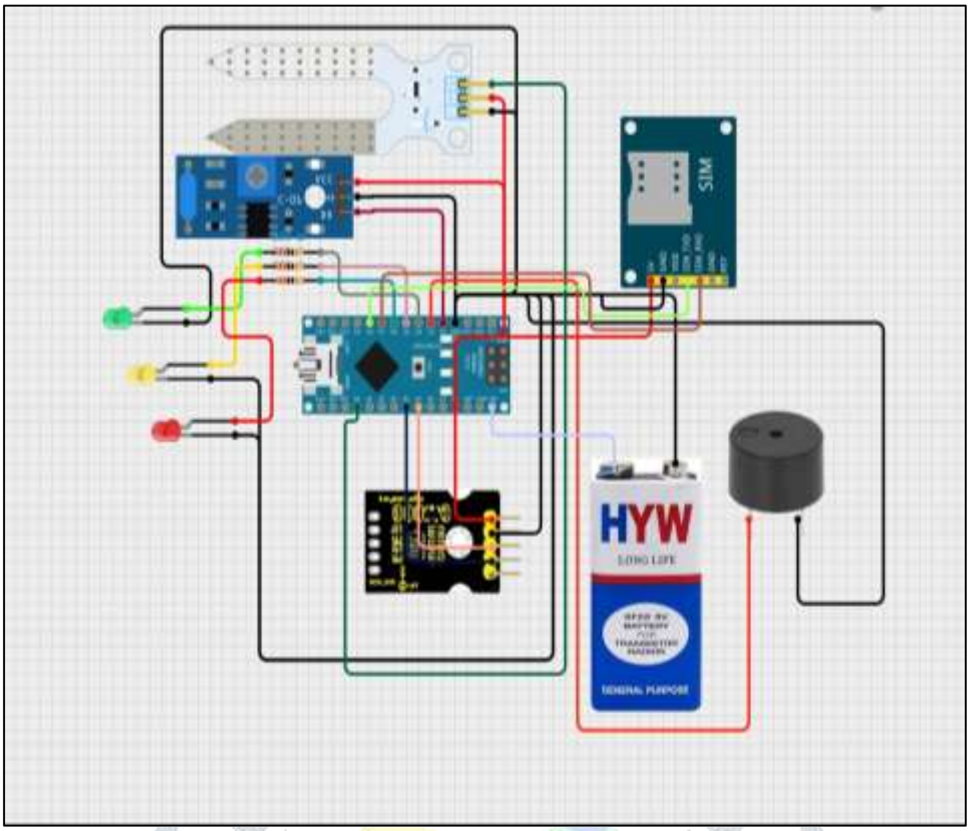

Fig.2-: Circuit Diagram

5. RESULTS & ANALYSIS

1. Pycnometer Test: The specific gravity value of the soil we collected from the site is displayed here. Specific gravity = 2.57 Kg/m3 A soil's specific gravity of 2.57 might help clarify some of its properties. A soil with a specific gravity of 2.57 can possess specific characteristics.

2. Direct Shear: The test results yield the following values: Internal friction angle = 19 degrees The soil's resistance to shearing, or sliding along a plane, is obtained by cohesion = 0.36 Kpa, the angle of internal friction. It is the angle at which the soil begins to slide and is often measured in degrees.

Here, the soil has an angle of internal friction of 19 degrees, which is low and suggests the soil is weak. The cohesion of the soil is a measure of the strength of the soil when it is subjected to stress, even without shearing. It is the stress required to cause failure of the soil. In this case, we have a soil with cohesion of 0.36 kPa, which is also a low value and represents a weak soil that is likely to fail. Using the test results a graph is drawn of normal stress vs. shear stress. Shear Stress vs Normal Stress graph Shear stress and normal stress. Shear Stress vs Normal Stress Graph.

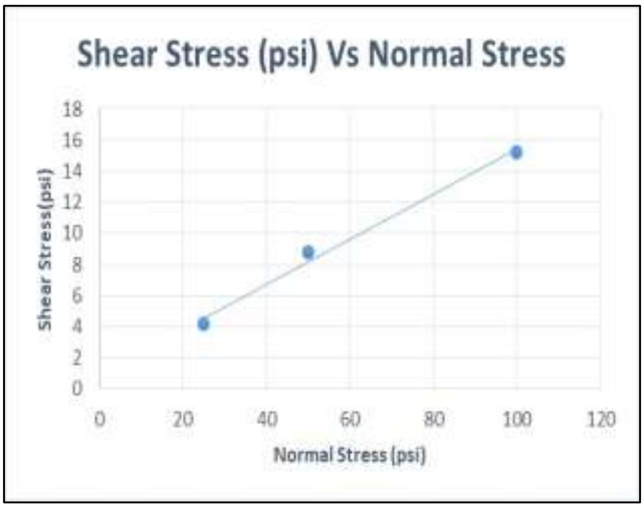

Chart-1: Shear Stress Vs Normal Stress Curve

3. Standard Proctor Test: We can determine the Optimum Moisture Content (0MC) of the soil removed from the site based on the test results. Optimum Moisture Content is 17%. If a certain type of soil has an OMC of 17%, it indicates that the soil has the highest level of compatibility and the lowest level of permeability when its moisture content is at 17% of its dry weight after being compacted to the maximum density using the standard compaction test. Even if the OMC of a soil is deemed as the best for a specific engineering purpose, it may still be prone to landslide due to factors such as extreme rainfall and earthquakes.

Fig.4: Standard Proctor

- **Rainfall Intensity:** In the rainfall simulation, we observed several key environmental factors related to the possibility of a landslide. The peak rainfall intensity was 40 mm per hour, which is a heavy rainfall that can quickly saturate the soil.
- **Soil Moisture Content:** Therefore, the soil moisture content increased substantially from 15% to 65%. This is especially significant as it is approaching the critical level at which the soil loses its bearing capacity, making slope failures more likely.
- **Tilt Sensor**: Apart from soil moisture, the movement of the ground was also measured with a tilt sensor. This sensor detected a slope change of 3.2°, suggesting initial signs of movement. This is a critical alert as angular changes can indicate potential soil instability.
- **Threshold Alerts:** The system was configured with safety thresholds to trigger alerts. These were set when the tilt angle was greater than 2.5° or when the moisture content was greater than 65%.

Both these thresholds were met in the simulation, showing the monitoring system's ability to detect dangerous situations. The high intensity of rainfall combined with increased soil moisture and the change in slope angle suggests a potential for a landslide to happen if not monitored closely. In summary, the results highlight the need for real-time monitoring and early warning systems for disaster risk reduction and mitigation.

Table-1: Results & Analysis

Parameter	Value	Threshold	Status
Rainfall Intensity	35mm/hr	40mm/hr	High Risk
Soil Moisture	65%	75%	Critical
Tilt Angle	3.2^0	2.5°	Danger
Response Time	5-10 sec	-	Good

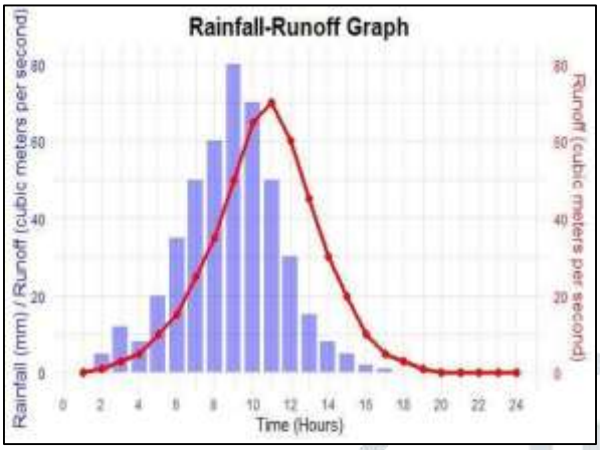

Chart-2: Rainfall Intensity

- Peak rainfall = 40 mm/hr
- Landslide risk increases after heavy Rainfall

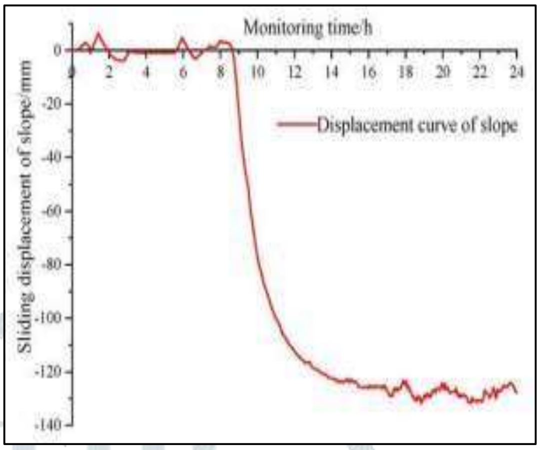

Chart-3: Land displacement over time

Critical tilt = 2.5°
Observed = 3.2° (danger zone)

7. CONCLUSION

1. We may conclude from the experimental model's procedure that the soil fails at 75% moisture content. To avoid the slopes of collapse, the buzzer was set to trigger an alarm at 65% moisture content.

2. The buzzer functioned at the given value and gave prior information before the landslide according to our model working. So the above project helps us to determine the possible landslides in the hills.

3. The goal of this project is to monitor the moisture content of the soil as well as fissures developed in the soil strata due to soil movement, which will provide us with a threshold value for triggering the landslide alarm.

4. These readings can be given to the local authorities for monitoring and for timely evacuation and mitigation. This will help in minimising economic loss and human lives. So, in this project we will try to improve the existing landslide determination systems.

5. After studying every paper, we concluded to the judgment that acceleration sensors, soil moisture sensors, and other sensors can be utilized to detect slides and strengthen early warning processes.

8. IMPLEMENTATION

The project can be implemented in the real environment in the following way.

- Supplying a module that will be buried 1.5–2 meters deep and includes a 3-axis gyroscope and two soil moisture sensors.
- These units can be positioned in the red zone between 500 and 1 km apart. This is to address the recent trend of cloud bursts in an exact location.
- Network-Based Deployment is deploying multiple sensor modules at a distance of 500 meters to 1 kilometre in the vulnerable (red zone) area to cover a large area and enhance the detection of localised events such as cloud burst.
- Real-Time Data Monitoring System is integrating all sensors with the Arduino-based microcontroller system which will monitor the real-time data like soil moisture, vibration and tilt.
- Buzzer & LED-based Alert System will issue alarm and evacuation signal for local warning in the form of Buzzer and LED when the sensor readings go beyond a certain threshold.
- Remote Monitoring & Communication use GSM/IoT modules for transmitting alarms and sensor data to control rooms for decision making and management of the event.

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